Gains to Trade in the Market for Distressed Debt[1]

Authors: Rohit Prasad, Professor, MDI Gurgaon

Yogesh B. Mathur, Independent Consultant

Email (corresponding author): rohit@mdi.ac.in

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Abstract

In this paper we build a theoretical model to analyze the gains to trade in the secondary market for distressed debt. In particular we focus on three critical variables – the proportion of cash and security receipts used in the transaction; the relative returns of the default bankruptcy process and the distressed debt market the relative risk aversion parameters of buyers and sellers of distressed debt. We bootstrap the model to the growth of the Indian distressed debt market since 2002, and make policy recommendations.

Key words: Asset Reconstruction Companies, Non-performing Assets, Insolvency and Bankruptcy Code, Debt Recovery.

[1] The generous inputs of Professor Allen Broughton, Professor of Mathematics-Rose-Hulman Institute of Technology brought significant clarity to the understanding of the ARC's decision problem. The usual disclaimer applies.
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1. Introduction

The economic slowdown resulting from the corona virus has resulted in a sharp increase in financial stress all over the world. The banking and non-bank finance sectors were already under significant amounts of stress even before the health crisis. For instance, as of October 2019, Russia and India had a gross NPA-loan ratio of 10.7 percent and 10.3 percent respectively. In a report dated October 2019, the IMF had stated that 'in a material economic slowdown scenario, half as severe as the global financial crisis, corporate debt-at-risk (debt owed by firms that are unable to cover their interest expenses with their earnings) could rise to $19 trillion—or nearly 40 percent of total corporate debt in major economies—above crisis[2] levels' (IMF 2019). Hence, in a post-Covid world one can expect sharply increased levels of financial distress resulting in an even higher accumulation of NPAs.

Financial pressures on banks result in a slowing down of the flow of credit. A case in point is India where the growth of bank credit has been as low as 6.19% in recent times with an upturn in 2017-18 being followed by a sharp downturn in 2019-20 (India Macro Advisors 2020). The low growth of credit

[2] 2008 financial crisis

exacerbates the challenges faced by business and in turn inhibits the recovery of the balance sheets of banks.

Across the world there is a raft of policies that needs to be operationalized to address the ongoing crisis including macro-prudential policies with regard to capital flows, and greater transparency, independence, and disclosure in financial practices. The creation of an efficient bankruptcy law and a well-functioning market for distressed debt is an important piece of the policy puzzle, especially in emerging economies. Even a country like India that has undergone a process of economic liberalization since 1991 notified a bankruptcy law, the Insolvency and Bankruptcy Code, only in 2016. During the nascent period of the promulgation of the law, resolution mechanisms continue to be weak resulting in poor recovery of dues for banks. The spotlight, thus, turns to the role of Asset Reconstruction Companies (ARCs), which form a channel for the sale of distressed debt without recourse to the bankruptcy process, although in most jurisdictions ARCs can enter the market after the declaration of bankruptcy as well.

In India, ARCs have been in existence since the promulgation of SARFESI Act in 2002. Although NPA sale transactions between banks and ARCs have resulted in an accumulation of assets of approximately Rs 1 lakh crores (approximately USD 15.4 billion) under the management of ARCs, this represents only 10% of the total stock of NPAs. Further the growth of assets under management of ARCs is slowing while the growth of NPAs is rising.

Hence, an analysis focusing on the incentives of the participants in the market for distressed debt is the need of the hour. This paper is a step in that direction. We analyze the gains to trade in the ARC market with the aim identifying the levers that would enable this market to function. For this purpose, we build a theoretical model which examines the decision of banks to go for a bankruptcy process or to opt for the sale of NPAs to ARCs; and the decision of ARCs to participate in the market in order to get the desired return on their capital. The model is applicable to any market where transactions in distressed debt are permissible.

To test the practical applicability of the model, we bootstrap the model with respect to historical data in the Indian ARC industry, and make policy recommendations that would enable this market to function with greater vigour. While there have been economic studies examining how the distressed debt market can be activated to ease the stress on the banking system in specific jurisdictions- for instance in Japan (Kazunari and Singh 2004), and in the USA (Altman 2014) - to the best of our knowledge, this paper represents the first attempt to build a theoretical model which can be bootstrapped to various jurisdictions in order to generate specific policy recommendations.

An important variable in the analysis is the relative proportion of cash and securities used to pay the bank. In a scenario of complete financial markets this proportion would be irrelevant. However, given the reality of incomplete markets, this variable assumes critical importance.

The rest of the paper is divided as follows: section 2 introduces our theoretical model and presents its results, section 3 discusses the enablers of a feasible market, section 4 discusses a variant of the model that incorporates a crucial regulatory feature related to the proportion of cash in the transaction between the bank and the ARC; Section 5 bootstraps the model to historical data in the Indian market; section 6 provides recommendations and concludes.

2. The Model

Gains to trade in the distressed debt market exist for several reasons: collection is a core competence of ARCs, avoidance of the label of bankruptcy saves a significant amount of value for the corporate debtor, securitization of NPAs enables risk sharing, and the specialization of ARCs in the reconstruction of assets increases the chances of recovery of the stressed company. However, several factors dampen these gains to trade. These include the increase in the cost of capital, incompleteness of financial markets, rules related to the eligibility of market players as well the prescribed minimum cash component of transactions. There have been drastic regulatory changes in the recent past with the prescribed minimum cash investment in buying NPAs swinging from 5% to 90%.

In this section, we create a theoretical model examining the viability of the market for distressed debt by examining the determinants of the decision of banks and Asset Reconstruction Companies (ARCs) to participate in the market. But first we need to understand the functioning of the ARC market: Let the written down value an asset in the books of the bank be Rs 1. The bank sells this asset to the ARC at a deal value given by Rs. C. The ARC holds this NPA in a trust and issues security receipts against this asset. A certain proportion, k, of the transaction with the bank is recovered by the bank in the form of cash, the remainder (1-k) is held in the form of security receipts and shown as an investment in the books of the bank. The ARC charges management fees for every rupee of security receipts held by the bank. The cash component of the transaction translates into security receipts held by the ARC in its own name.

2.1. The Bank's Decision

A bank's decision to recover its NPAs via the IBC route or, alternatively, through a sale to an ARC is a function of the following key factors: expected present value of recovery from IBC, deal value in the ARC transaction, the proportion of upfront cash from sale to ARC, expected present value of the returns from the reconstructed NPA, and the management fee charged by the ARC. Other factors, such as the reduction in the volume of NPAs, are also important. These nuances are introduced after analyzing the base model.

We assume that for a bank to sell NPAs to an ARC, the value it derives from the payouts under the IBC process should be lower than the returns from the ARC. We assume that the book value of the NPA sold by the bank is Rs 1 and use the following notation to model the bank's decision:

C: deal value (this is the value at which the NPA worth Rs 1 is sold by the bank to the ARC and paid out with a combination of cash & SRs)

k: proportion of cash in the NPA sale deal

f: sum of present value of annual management fee to ARC

S_I: *present value of recovered value of the asset when the bank goes for the IBC process*

Note that S_I is paid in cash while the returns from the ARC come in a mixture of cash and security receipts.

The value to the bank if it goes to the ARC is composed of two components a cash component – kC, and a security receipts component (1-k)C. The value attached by the bank to the security receipts component depends on its expectations of the performance of the ARC and the broader economy. In a buoyant economy the bank would assess its return from the overall deal – cash and security receipts, as equal to the deal value, C. On the other hand, in a depressed economy, the bank would assess its value as restricted to the cash component, kC. In normal times, the bank would assess its value as somewhere in between kC and C.

We express the bank's valuation of the deal as k^nC where $n \in [0, 1]$ represents the bank's level of risk aversion. When 'n' is equal to 1, the bank is completely risk averse and only ascribes value to the cash component of the deal. When 'n' is equal to 0, the bank is optimistic about economic prospects and believes that it will recover the entire cash value of the deal[3].

While the returns from the IBC process, and the ARC process depend on some distinct factors, both are similarly impacted by macroeconomic conditions. For instance, if conditions in the steel industry improve, both the IBC process and the ARC process would yield higher returns. Hence we make the deal value, C, a function of the payout from the IBC, S_I, which you will recollect is paid in cash. We set $C = bvS_I = PS_I$ where b is the 'bankruptcy avoidance premium' from the ARC process on account of the distressed debt <u>not</u> going through the bankruptcy process if it is sold to the ARC, and v is the additional value generated by the ARC through securitization.

The labelling of a company as bankrupt has a chilling effect on its already dim prospects. Vendors, customers, and employees start having second thoughts about associating with this company. Certain inhibiting rules get triggered – for instance, the rule barring an

[3] When 'n' is less than 0, the bank is so optimistic about economic prospects that it believes it will recover more than the value of the deal on account of its holding of security receipts. We do not consider this case.

infrastructure company from accepting new orders. Hence, b ≥ 1. If b =1, the ARC enjoys no premium on account of having avoided the bankruptcy process.

The ARC also engages in a process of securitization of the debt. Securitization takes two forms – first,the debt is split into smaller units and sold ; second, the debt is bundled with other debt instruments to create a diversified portfolio. The additional value generated by the ARC through securitization is represented by v. Hence, v ≥ 1. If v =1, the ARC creates no value on account of securitization. The total premium created by the ARC, P= bv - a combination of the bankruptcy avoidance premium, b, and the securitization and reconstruction premium, v. We assume P >1.

Finally, the management fees paid by the bank at the rate of Rs. f on every rupee of security receipt held also has to be considered. The management fees paid = (1-k)Cf

Hence, the surplus generated for the bank by the ARC is given by

$$S_B \equiv k^n C - (1-k)Cf - S_I$$

The bank will go in for the deal as long as

$$k^n C - (1-k)Cf - S_I \geq 0$$

Substituting C = PS_I we get

$$k^n PS_I - (1-k)PS_I f - S_I \geq 0$$

This implies

$$B \equiv k^n - (1-k)\, f - 1/P \geq 0$$

Note B is a function of k, f, and P.

Lemma 1: *For all* $n \in [0,1], P > 1, f > 0,$ *there exists a unique threshold* $k_B^* \in [0,1)$*, such that the bank will transact if and only if* $k \in [\, k_B^*, 1]$.

Proof: Note $\frac{\partial B}{\partial k} = nk^{n-1} + f > 0$ for all points $n \in (0,1] X\, k \in [0,1]$. At $n = 0$, $\frac{\partial B}{\partial k} = f > 0$. This implies for all $n \in [0,1]$, B is a strictly increasing function of k that is maximized at k =1.

Further, $\frac{\partial^2 B}{\partial k^2} = n(n-1)k^{n-2} \leq 0$ for all $n \in [0,1]\, X\, k \in [0,1]$. If $n = 0$ or $n = 1$ or $k = 0$, $\frac{\partial^2 B}{\partial k^2} = 0$. Thus B is a concave function of k with strict concavity at $n \in (0,1) X\, k \in (0,1]$.

For all $n \in (0,1]$, $\mathrm{B(k=0)} = -\left(\mathrm{f} + \frac{1}{P}\right) < 0$, and $\mathrm{B(k=1)} = 1 - \frac{1}{P} > 0$. By the intermediate value theorem, B must achieve at least one zero in the interval $k \in (0,1)$.

Suppose there are two zeros, k_1 and k_2[4]. By Rolle's Theorem, the slope of B must achieve a zero at a point c in between k_1 and k_2, i.e. there exists $c \in (k_1, k_2)$ such that $\frac{\partial B}{\partial k}(c) = 0$.

For all $n \in (0,1)$, $\frac{\partial B}{\partial k}$ is a strictly decreasing function of k. Hence, $\frac{\partial B}{\partial k}$ must be strictly positive in the interval $k \in [0, c)$ and strictly negative in the interval $k \in (c, 1]$. Since $c < k_2 < 1$ this implies B has a strictly negative slope in the interval $k \in [k_2, 1]$. Thus $B(k=1) < B(k=k_2) = 0$, a contradiction.

For $n = 1$, $\frac{\partial B}{\partial k} = 1 + f > 0$, for all k. Hence there is no $c \in (k_1, k_2)$ such that $\frac{\partial B}{\partial k}(c) = 0$, a contradiction.

For n =0, $\frac{\partial B}{\partial k} = 1 + f > 0$. If $\mathrm{B(k=0)} = 1 - \mathrm{f} - \frac{1}{\mathrm{P}} \leq 0$, the same argument as for $n = 1$ shows that B achieves a zero at a unique $k_B' = 0$.

At n =0, suppose $\mathrm{B(k=0)} = 1 - \mathrm{f} - \frac{1}{\mathrm{P}} > 0$. Note $\mathrm{B(k=-\infty)} = -\infty$. The same argument as for $n = 1$ shows there exists a unique $\mathrm{k_B'} \in (-\infty, 0)$ such that $\mathrm{B(k_B')} = 0$. Let $k_B^* \equiv \max\{k_B', 0\}$[5]. Since for all $n \in [0,1]$, B is a strictly increasing function of k, a transaction will take place if $k \geq k_B^*$.

The conclusion follows.

Lemma 2: *For* $n \in (0,1]$, $(i)\ \frac{\partial k_B^*}{\partial n} > 0\ (ii) \frac{\partial k_B^*}{\partial f} > 0 (iii) \frac{\partial k_B^*}{\partial P} < 0$. *For n=0, if* $1 - f - \frac{1}{P} < 0, (i)\ \frac{\partial k_B^*}{\partial n} > 0\ (ii) \frac{\partial k_B^*}{\partial f} > 0 (iii) \frac{\partial k_B^*}{\partial P} < 0$. *If* $1 - f - \frac{1}{P} = 0, (i) \frac{\partial k_B^*}{\partial f} > 0 (ii) \frac{\partial k_B^*}{\partial P} < 0$. *If* $-f - \frac{1}{P} > 0, \frac{\partial k_B^*}{\partial f} = \frac{\partial k_B^*}{\partial P} = 0$. *For n=0,* $1 - f - \frac{1}{P} \geq 0$, k_B^* *increases with* n *but discontinuously.*

Proof: $\frac{\partial B}{\partial n}(k,n)$ is a continuous function at all $(k,n) \in [0,1]X[0,1]$ except at the points $(0,n)$ where $k = 0\ and\ n \in (0,1)$. $\frac{\partial B}{\partial k}(k,n)$ is a continuous function at all $(k,n) \in [0,1]X[0,1]$[6]. This implies that B is continuously differentiable at all $(k,n) \in [0,1]X[0,1]$ except at the points $(0,n)$ where $n \in (0,1)$.

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[4] The skeleton of this proof was provided by Professor Allen Broughton in an email exchange.

[5] Note, $k_B^* = k_B'$, unless $(i) n = 0\ and\ (ii) 1 - \mathrm{f} - \frac{1}{\mathrm{P}} > 0$.

[6] For $k = 0, n \in (0,1), \frac{\partial B}{\partial k} = \infty$, thus B has an improper derivative at $k = 0, n \in (0,1)$.

From Lemma 1, for $n \in (0,1)$, $k_B^* = k_B' \in (0,1)$ at all $f > 0, P > 1$. Hence B is continuously differentiable at all (k_B^*, n) where $n \in (0,1]$.

For all $n \in (0,1]$, $k \in (0,1)$, B is a strictly increasing function of k[7]. Hence $\frac{\partial B}{\partial k}(k_B^*) > 0, i.e.$ B is a strictly increasing function of k at the point $B(k) = 0$. Thus the necessary conditions of the implicit function theorem are satisfied for the equation $B(k, n, P) = 0$. Therefore,

$$\frac{\partial k_B^*}{\partial n} = -\frac{\frac{\partial B}{\partial n}}{\frac{\partial B}{\partial k_B^*}} = -\frac{{k_B^*}^n \log k_B^*}{n{k_B^*}^{n-1} + f}$$

Since $0 < k_B^* < 1$, ${k_B^*}^n \log k_B^* < 0, n{k_B^*}^{n-1} + f > 0$, and $\frac{\partial k_B^*}{\partial n} > 0$.

For (ii) and (iii) note $\frac{\partial B}{\partial f} = -(1-k) < 0$, for all $k \in [0,1)$, and $\frac{\partial B}{\partial P} = \frac{1}{P^2} > 0$ for all $k \in [0,1]$. The conclusion follows in the same manner as for (i).

For n=0, there are three cases (i) $1 - \mathrm{f} - \frac{1}{\mathrm{P}} < 0$ (ii)$1 - \mathrm{f} - \frac{1}{\mathrm{P}} = 0$, and (iii)$1 - \mathrm{f} - \frac{1}{\mathrm{P}} > 0$

$1 - \mathrm{f} - \frac{1}{\mathrm{P}} < 0$ implies $k_B^* = \mathrm{k}_\mathrm{B}' \in (0,1)$. B is a continuously differentiable function of n at $k = k_B^*$. B is also a continuously differentiable function of f, P for all $f > 0, P > 1$. Further, $\frac{\partial B}{\partial k_B^*} = f > 0$. Hence the implicit function theorem can be applied to show (i) $\frac{\partial k_B^*}{\partial n} > 0$ (ii) $\frac{\partial k_B^*}{\partial f} > 0$ (iii) $\frac{\partial k_B^*}{\partial P} < 0$.

For all f, P such that $1 - \mathrm{f} - \frac{1}{\mathrm{P}} = 0$, $k_B^* = \mathrm{k}_\mathrm{B}' = 0$. B is no longer a continuously differentiable function of n since $\frac{\partial B}{\partial n} = -\infty$ at $n = 0$, but $\frac{\partial B}{\partial n}$ is indeterminate for $n \in (0,1)$. However, we know that for all $n \in (0,1]$, $k_B^* = \mathrm{k}_\mathrm{B}' \in (0,1)$. k_B^* thus increases as n increases but discontinuously.

However, B is a continuously differentiable function of f, P for all $f > 0, P > 1$, and $\frac{\partial B}{\partial k_B^*} = f > 0$. Hence, (i) $\frac{\partial k_B^*}{\partial f} > 0$ (ii) $\frac{\partial k_B^*}{\partial P} < 0$.

For all f, P such that $1 - \mathrm{f} - \frac{1}{\mathrm{P}} > 0$, $k_B' < k_B^* = 0$. Hence $\frac{\partial k_B^*}{\partial f} = \frac{\partial k_B^*}{\partial P} = 0$. As before, k_B^* increases as n increases but discontinuously.

This completes the proof.

[7] Indeed this is true for $n \in [0,1]$.

From Lemma 2, k_B^* acquires its maximum value at n=1. When n=1, we find k_B^* by setting $B = (k_B^*)^1 - (1 - k_B^*)f - \frac{1}{P} = 0$. This implies $max\ k_B^* = \frac{\frac{1}{P}+f}{1+f} < 1$. No matter how risk averse the bank is if the cash proportion is greater than $\frac{\frac{1}{P}+f}{1+f}$ the bank will be willing to transact.

Recall $k_B^* \equiv \max\{k_B', 0\}$. From Lemma 1 and Lemma 2 we can conclude $k_B^* \in [0, \frac{\frac{1}{P}+f}{1+f}]$.

2.2. The ARC's Decision

The ARC's decision to purchase NPAs from banks is a function of the following key factors: deal value in the ARC transaction; proportion of cash in the deal; expected present value of the returns from the reconstructed NPA; and the management fees.

The returns to the ARC come in a mixture of management fees and security receipts. The cash component of the deal, kC, represents security receipts held by the ARC in its own name. (1-k)C represents security receipts held in the bank's name, for which the ARC earns management fees given by (1-k)Cf.

The value attached by the ARC to the security receipts component depends on its expectations with regard to its own performance and that of the broader economy. In a buoyant economy the ARC would assess its return as kC, i.e. reflecting its belief that the deal value represents the true present value of security receipts. On the other hand, in a depressed economy, the ARC would assess its value as restricted to the management fees. In normal times, the ARC would assess the value of its security receipts as somewhere in between kC and 0.

We express the ARC's valuation of its security receipts as $k^m C$ where m, which represents the ARC's level of risk aversion. We assume $m \in [1, \infty]$. When 'm' is equal to 1, the ARC is optimistic about economic prospects and believes that they will recover the entire value of the security receipts they hold. As 'm' tends to infinity, the ARC is completely risk averse and only ascribes value to the management fees component of the deal. When 'm' is a finite number greater than 1, the ARC believes it will recover a positive amount from the security receipts, but not the entire value.

As before, setting C = PS_I, the ARC's return on kC, the capital deployed, is given by

$$R_A = (k^m PS_I + (1-k) PS_I f)/k\, PS_I = (k^m + (1-k)f)/k$$

The net return of the ARC, NR_A, is given by the return minus Z, the cost of capital of the ARC. We assume Z >1.

$A = \frac{k^m \ + \ (1-\mathrm{k})\mathrm{f}}{k}$ - Z = $\frac{k^m \ + \ (1-\mathrm{k})\mathrm{f} \ -\mathrm{Zk}}{k}$.

Let $A' \equiv k^m - (Z+f)k + f$

Lemma 3: *For all* $m \geq 1, f > 0, Z > 1,$ *there exists a unique threshold* $k_A^* \in (0,1)$, *such that the ARC will transact if and only if* $k \in [\,0,\ k_A^*\,]$.

Proof: At k = 0, $A = \infty$. At all other values of $k \in [0,1]$, A is a finite quantity. Thus A *is maximized at k = 0*.

A deal will go through if the net return of the ARC is positive, i.e. if:

A = $\frac{k^m \ + \ (1-\mathrm{k})\mathrm{f} \ -\mathrm{Zk}}{k} \geq 0$

Note $k \geq 0$, i.e. the denominator is always positive. Thus the ARC will go in for the deal if and only if the numerator $A' \geq 0$[8]. Given $Z, f,\ A'$ is a continuously differentiable function of m and k, $A': [1,\infty) X [0,1] \rightarrow R$. For all $m \geq 1, A'\,(k=0) = f > 0,$ and $A'\,(k=1) = (1-Z) < 0$. By the intermediate value theorem, A' achieves a zero in the interval $k \in (0,1)$.

Further, $\frac{\partial^2 A'}{\partial^2 k} = m(m-1)k^{m-2} \geq 0$ for all $k \in [0,1]$ with strict inequality if $m > 1$ and $k > 0$. Thus A' is convex and has no inflection points in the interval $k \in [0,1]$. It follows that in the $(0,1)$ interval A' achieves a zero at exactly one point[9][10].

Lemma 4: For $\boldsymbol{all\ m} \geq \mathbf{1}$, (i) $\frac{\partial k_A^*}{\partial m} < 0$ *(ii)* $\frac{\partial k_A^*}{\partial Z} < 0$*(iii)* $\frac{\partial k_A^*}{\partial f} > 0$.

Proof: Recall,

$$A' = k^m - (Z+f)k + f$$

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[8] For both the bank and the ARC we assume that in case of indifference between transacting and not transacting, the transaction will take place.

[9] The rigorous proof of this rather intuitive assertion is similar to the proof of Lemma 1 and proceeds as follows: Suppose there are two zeros k_1 and k_2. By Rolle's Theorem, the slope of A' must achieve a zero at a point c between k_1 and k_2. Since the slope is an increasing function of k, it must be negative in the interval $k \in [0,c)$ and positive in the interval $k \in (c,1]$. Since $c < k_2 < 1$ this implies A' has a positive slope in the interval $k \in [k_2,1]$. Thus $A'(k=1) > A'(k=k_2)$ =0, a contradiction.

[10] At $k = \infty, A'$ is unbounded. Thus the function has exactly two zeroes, one in the $(0,1)$interval and one in the interval $(1,\infty]$.

$\frac{\partial A'}{\partial m}(k,m)$ is a continuous function at all $(k,m) \in (0,1]X[1,\infty)$. $\frac{\partial A'}{\partial k}(k,m)$ is a continuous function at all $(k,m) \in [0,1]X[1,\infty)$[11]. By Lemma 3, $k_A^* \in (0,1)$. This implies that B is continuously differentiable at all $(k_A^*, m) \in (0,1]X[1,\infty)$.

As shown in the proof of Lemma 3, for all $m \in R, A'$ takes on a positive value at k=0 and a negative value at k =1. Further, for $m \geq 1, \frac{\partial^2 A'}{\partial^2 k} \geq 0, i.e. \frac{\partial A'}{\partial k}$ is a non-decreasing function of k. Suppose $\frac{\partial A'}{\partial k}(k_A^*, m) \geq 0$ at $k_A^* \in (0,1)$. Then $\frac{\partial A'}{\partial k}(k_A^*, m) \geq 0$, for all $k \in (k_A^*, 1]$. This implies $A'(k=1) \geq A'(k=k_A^*)$ =0, a contradiction[12]. Hence for all $m \geq 1$, $\frac{\partial A'}{\partial k}(k_A^*, m) < 0.$[13]

This implies that the implicit function theorem can be applied and

$$\frac{\partial k_A^*}{\partial m} = -\frac{\frac{\partial A'}{\partial m}}{\frac{\partial A'}{\partial k}} = \frac{{k_A^*}^m \log k_A^*}{m{k_A^*}^{m-1} - (\mathrm{Z}+\mathrm{f})}$$

Recall, $k_A^* \in (0,1)$. Therefore ${k_A^*}^m \log k_A^*$ is always negative. Hence $\frac{\partial k_A^*}{\partial m}$ is negative if and only if $\frac{\partial A'}{\partial k}(m, k_A^*) = m{k_A^*}^{m-1} - (Z+f) < 0$. We have already established that A' is negatively sloped at (m, k_A^*). The result follows.

To prove parts (ii) and (iii), note $\frac{\partial A'}{\partial Z}$ is a continuous function of Z, and $\frac{\partial A'}{\partial f}$ is a continuous function of f. $\frac{\partial A'}{\partial Z} = -k < 0$, and $\frac{\partial A'}{\partial f} = -(1-k) < 0$ for $k \in (0,1)$. The results follow.

From Lemma 4, k_A^* achieves its maximum value at $m=1$, and its minimum value at $m = \infty$. At $m = \infty$, $k_A^* = \frac{f}{Z+f,}$ while at $m = 1$, $k_A^* = \frac{f}{Z+f-1,}$. Thus $k_A^* \in [\frac{f}{Z+f}, \frac{f}{Z+f-1}]$.

3. Market Feasibility

A market transaction will take place if both the bank and the ARC want to transact. For any n, f, P, we compute k_B^* . This is the minimum cash proportion acceptable to the bank. Similarly, for any m, f, Z, we compute k_A^* . This is the maximum cash proportion acceptable to the ARC. Note from Lemma 1 and Lemma 3, a unique k_A^* and k_B^* exist. For market feasibility, we require $k_A^* \geq k_B^*$,

[11] For $k=0, n \in (0,1), \frac{\partial B}{\partial k} = \infty$, thus B has an improper derivative at $k=0, n \in (0,1)$.

[12] Note that this assertion is true despite the fact that for $m > Z+f, \frac{\partial A'}{\partial k}(k=1) > 0$.

[13] The skeleton of this proof was provided from Professor Broughton in an email exchange.

i.e. the maximum cash-proportion at which the ARC will transact must be greater than or equal to k_B^*, the minimum cash proportion acceptable to the bank. The minimal condition of feasibility is $k_A^* = k_B^*$. Given the sluggish nature of the market for distressed debt, particularly in a recession, we focus on the minimal condition of feasibility in this paper.

A minimally feasible market situation can be graphically shown as follows:

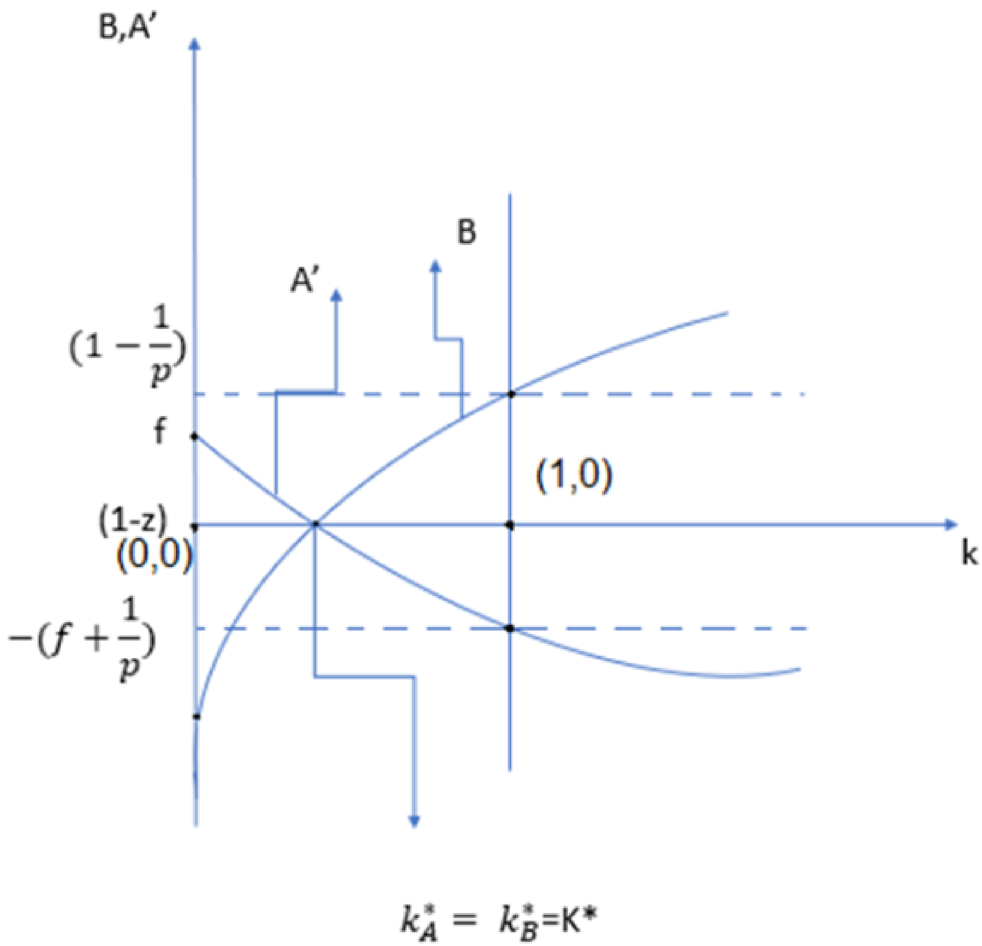

Figure 1: Minimally Feasible Market

Note if $k_B^* > \frac{f}{Z+f-1}$, no transaction is possible since $k_A^* \leq \frac{f}{Z+f-1}$. Also, if $k_B^* < \frac{f}{Z+f}$, then a transaction is possible no matter how risk averse the ARC may be since since $k_A^* \geq \frac{f}{Z+f}$.

Ideally we would like to derive conditions on n, f, P, m, Z that would characterize a minimally feasible market. However, that has not proved to be possible. Hence we derive necessary and sufficient conditions under 4 conditions: (i) n=0, and m=1, i.e. both the bank and the ARC are sure of realizing the full return from the security receipts, (ii) $n = 0, and\ m = \infty$, i.e the bank is sure of realizing the full return from the security receipts, but the ARC assesses the return from the security receipts it holds as nil. (iii) n=1, and m=1, i.e the ARC is sure of realizing the full return from the security receipts it holds, but the bank assesses the return from its security receipts as nil, and (iv) $n = 1, and\ m = \infty$, i.e. both the bank and the ARC assess the return from the security receipts they hold as nil.

At n=0, $k_B^* = \frac{Pf+1-P}{Pf}$. At n=1, $k_B^* = \frac{Pf+1}{Pf+P}$. At m=1, $k_A^* = \frac{f}{Z+f-1}$, and at $m = \infty$, $k_A^* = \frac{f}{Z+f}$. Note when deriving the value of k_A^* at $m = \infty$,we use the fact that $k_A^* \in (0,1)$ as derived in Lemma 3.

This gives us the following conditions characterizing a feasible market:

Table 1: Conditions Characterizing a Feasible Market

ARC	m=1	m =∞
Bank		
n = 0	f$(PZ - P) \leq (Z - 1 + f)(P - 1)$	$PfZ \leq (Z - f)(P - 1)$
n =1	$(Pf + 1)(Z - 1) \leq f(P - 1)$	$Pf(Z - 1) \leq -(Z + f)$

Note, given the assumptions that $P > 1, Z > 1, f > 0$ Case (iv) is impossible.

3.1. Equilibrium at Different Points of the Business Cycle

We analyze equilibrium at the crest and trough of the business cycle. This allows us to assess the possibility of mutual gains to trade in the secondary market for distressed debt at different points of the business cycle, especially at the trough where one expects the need for the secondary market for distressed debt to be intensified.

Our assumption is that the values of the relevant parameters at the crest and the trough differ as follows:

Table 2: Values of Parameters at Crest and Trough of Business Cycle

Parameter	Value at Crest compared to Value at Trough is…	Comments
n	Lower	Bank more risk averse at trough
P	Higher	Recall P is composed of 2 parts: b, the bankruptcy avoidance premium which is lower at crest, and v, the perceived value generated by the ARC through securitization, which we assume is higher at the crest. Our assumption is that the combined effect of these two variables is to make $P_{crest} \geq P_{trough}$
f	same	Usually stays fixed across business cycle. Relatively insignificant.
Z	lower	Easier to obtain capital at crest

Lemma 5: *Assume there is a minimally feasible market situation at the crest, and that the bank becomes more risk averse when going from the crest to the trough. Then the ARC must be less risk averse at the trough than at the crest to enable a transaction to take place.*

Proof: By assumption $n_{trough} > n_{crest}$.

From Lemma 2, this implies $k^*_{B,trough} > k^*_{B,crest}$. Thus the ARC has to be tolerant of a higher cash proportion at the trough than at the crest, i.e. $k^*_{A,trough} > k^*_{A,crest}$. From Lemma 4, this implies $m_{trough} < m_{crest}$.

Thus, even though the bank behaves pro-cyclically, becoming more risk averse at the trough, the ARC has to act counter-cyclically, becoming less risk averse at the trough, in order to enable the transaction to happen. This conclusion is derived under the assumption that $P_{crest} \geq P_{trough}$. However, if market players are able to predict the business cycle accurately, then $v_{crest} \leq v_{trough}$ and $P_{crest} < P_{trough}$. In this case, if the difference in the premium earned by the bank at the trough and the crest is high enough, it may outweigh the higher risk aversion of the bank. As a result it is possible that $k^*_{B,trough} < k^*_{B,crest}$, and the ARC can afford to take a pro-cyclical view. More precisely, a necessary condition for the ARC to be able to act pro-cyclically is that $\frac{v_{crest}}{v_{trough}} < \frac{b_{trough}}{b_{crest}}$.

In order to explore enabling factors for the transaction to happen we slightly modify the assumption on the bank's behavior. Either a. the bank sets expectations in a highly pro-cyclical way, or b. the bank sets expectations in a less pro-cyclical way. We loosely refer to the less pro-cyclical stance as 'counter-cyclical'. In both cases $n_{trough} > n_{crest}$. However, we assume that a pro-cyclical bank would be more bullish at the crest than a counter-cyclical bank, i.e. $(n_{crest})_{\text{pro-cyclical}} < (n_{crest})_{\text{counter-cyclical}}$. On the other hand, a pro-cyclical bank would be more bearish at the trough than a counter-cyclical bank, i.e. $(n_{trough})_{\text{pro-cyclical}} > (n_{trough})_{\text{counter-cyclical}}$. Under these assumptions we derive the following corollary in the same manner as Lemma 5.

Corollary 1: *To ensure market feasibility, the ARC can afford to be more risk averse at the crest with a pro-cyclical bank than with a counter-cyclical bank. However, the ARC is required to be less risk averse at the trough with a pro-cyclical bank than with a counter-cyclical bank.*

Given that a pro-cyclical bank is likely to generate more NPAs at the trough than a counter-cyclical bank, the market for distressed debt is subjected to a double whammy with a pro-cyclical bank – not only is there a greater volume of distressed debt that requires to be cleared, but the level of risk aversion expected of the ARC is also lower. On the other hand a counter-cyclical banking system generates fewer NPAs and allows a higher level of risk aversion on the part of the ARC at the trough.

4. A Variation of the Model Incorporating Accounting Benefits to the Bank

In the model above, the surplus of the bank was independent of compliance with the directives of the central bank with respect to attaining a minimum cash proportion in the transaction with the ARC. However, this omission masks an important aspect of reality. Central bank regulations require banks to fully provision for the NPA over a certain period of time (4 years in India), by showing an annual loss on the P&L statement and writing down the value of the NPA in the balance sheet by the same amount. If the NPA is sold then the bank avoids having to make this provision. This represents an important incentive for the bank to transact with the ARC. There is, however, a catch.

In India, and in some other jurisdictions as well, the central bank allows this benefit only if the transaction in the distressed debt incorporates a minimum cash proportion. In India this stipulated cash proportion has varied from 5% in the period 2006 to 2013 to as high as 90% in the period starting 2017.

Further, the central bank has been concerned about the high levels of discounts provided on even the written down value of the NPA. Hence, the relief on provisioning extends to the total value of the deal (denoted as C in our mathematical model) or the written down value, W, whichever is the lower.

To illustrate, if the book value of a two-year-old NPA is Rs. 100, $W = Rs. 50$, as Rs 25 would have been provisioned over each of the two years of its existence (assuming the provisioning period is 4 years). Suppose $C = Rs. 30$ and suppose that the minimum cash proportion stipulated by the central bank is satisfied. Then the bank only saves Rs. 30 on provisioning. It will continue to have to provision for the loss on the written down value of Rs. 50, i.e. Rs. 20. On the other hand, if $C \geq Rs. 50$, it will save the entire written down value of Rs. 50.

Denoting the stipulated minimum cash proportion as $\dot{k}$, the surplus to the bank, S_B, can be written as follows:

$S_{B1} \equiv k^n C - (1-k)Cf - S_I + \min\{W, C\}$, if $k \geq \dot{k}$, or

$S_{B2} \equiv k^n C - (1-k)Cf - S_I$, if $k < \dot{k}$

As before a transaction will occur if $S_B \geq 0.$ Dividing both sides of the equation by C we get the condition

$B_1 \equiv k^n - (1-k)f - \frac{1}{P} + \min\{\frac{W}{C}, 1\} \geq 0$ if $k \geq \dot{k}$,

$B_2 \equiv k^n - (1-k)f - \frac{1}{P} \geq 0$ if $k < \dot{k}$

Note that B_2 is identical to the condition we derived in the basic model without central bank regulation, and $B_1 = B_2 + \min\left\{\frac{W}{C}, 1\right\} \geq B_2$. For $n = 0$, $B_1(k=0) = 1 + \min\left\{\frac{W}{C}, 1\right\} - \frac{1}{P} - f$. For $n > 0$, $B_1(k=0) = \min\left\{\frac{W}{C}, 1\right\} - \frac{1}{P} - f$. For all $n \in [0,1]$, $B_1(k=1) = 1 + \min\left\{\frac{W}{C}, 1\right\} - \frac{1}{P} > B_1(k=0)$. $B_1$is a continuous, increasing, concave function of k that is maximized at k =1.

As before, let the threshold value of k that makes the bank indifferent between transacting and not transacting when $k < \dot{k}$ be k_B' and let $k_B^* = \min\{k_B', 0\}$. Similarly, let the threshold value of k that makes the bank indifferent between transacting and not transacting when $k \geq \dot{k}$ be k_B'' ,and let $k_B^{**} = \max\{k_B'', 0\}$. It is easy to see that $k_B^{**} = k_B''$, unless $B_1(k=0) > 0$.

Note that $\frac{\partial B_1}{\partial k} = \frac{\partial B_2}{\partial k} = nk^{n-1} + f > 0$ for all $n \in [0,1]$. Thus both B_1 and B_2 are parallel functions of k. Hence $k'' \leq k'$ with this inequality being a strict inequality if $\min\left\{\frac{W}{C}, 1\right\} > 0$.

We have already established the properties of k_B^* in Lemma 1 and Lemma 2. Since $B_1 = B_2 + \min\{\frac{W}{C}, 1\}$ the properties of k_B^{**} are similar. We state without proof the following Lemmas with respect to B_1.

Lemma 1'*: For all $n \in [0,1], P > 1, f > 0, W > 0, C > 0$ there exists a unique threshold $k_B^{**} \in [0,1)$, such that the bank will transact if and only if $k \in [k_B^{**}, 1]$.*

Lemma 2': *For $n \in (0,1], f > 0, P > 1, W \geq 0,$ and $C \geq 0$ such that $B_1(k=0) < 0$,then (i) $\frac{\partial k_B^{**}}{\partial n} > 0$, (ii) $\frac{\partial k_B^{**}}{\partial f} > 0$, (iii) $\frac{\partial k_B^{**}}{\partial P} < 0$ (iv) $\frac{\partial k_B^{**}}{\partial C} \geq 0$,* and (v) $\frac{\partial k_B^{**}}{\partial W} \leq 0$. *If $B_1(k=0) = 0$, then (i) $\frac{\partial k_B^{**}}{\partial f} > 0$, (ii) $\frac{\partial k_B^{**}}{\partial P} < 0$ (iii) $\frac{\partial k_B^{**}}{\partial C} \leq 0$,* and (iv) $\frac{\partial k_B^{**}}{\partial W} \geq 0$. *$k_B^{**}$ increases with n but discontinuously. If $B_1(k=0) > 0$, then $\frac{\partial k_B^{**}}{\partial f} = \frac{\partial k_B^{**}}{\partial P} = \frac{\partial k_B^{**}}{\partial C} = \frac{\partial k_B^{**}}{\partial W} = 0$. k_B^* increases with n but discontinuously.*

We now turn to a consideration of the threshold cash proportion of the bank in the model incorporating accounting benefits to the bank.

Lemma 6: *For* $n \in [0, 1], P > 1, f > 0, W > 0, C > 0, \dot{k} \geq 0,$ *there exists a unique threshold cash proportion,* $\ddot{k}_B$*, such that the bank will transact if and only if* $k \in [\ddot{k}_B, 1]$.

Proof: There are three cases to consider: (i) $k_B^* = k_B^{**} = 0$, (ii) $k_B^* > k_B^{**} = 0$, and (iii) $k_B^* > k_B^{**} > 0$.

Case (i) (see Figure 2) would occur only if $n = 0$, and $1 - f - \frac{1}{P} \geq 0$. In this case, $\ddot{k}_B = 0$ for all values of $\dot{k} \in [0, 1]$. If $\dot{k} = 0$, the bank will be in adherence with the central bank stipulations and will operate on the B_1 function , else it will operate on the B_2 function. For all values of $\dot{k}$ the bank will earn a strictly positive surplus.

In case (ii) (see Figure 3) for $0 \leq \dot{k} < k_B^*$, if the bank does not adhere to the minimum cash stipulation and operates on the B_2 function, it will make a loss. Hence the bank operates on the B_1 function and $\ddot{k}_B = \dot{k}$. For $\dot{k} = k_B^*$, if the bank chooses a cash proportion less than $\dot{k}$, it will operate on the B_2 function and make a loss. If it chooses a cash proportion more than $\dot{k}$, it will operate on the B_1 function and make a profit. Hence $\ddot{k}_B = k_B^*$, i.e. the bank will operate on the B_1 function and make a profit. If $\dot{k} > k_B^*$, $\ddot{k}_B = k_B^*$. The bank will operate on the B_2 function and be indifferent between transacting and not transacting.

In case (iii) (see Figure 4) for $0 \leq \dot{k} \leq k_B^{**}$, the bank will operate on the B_1 function and $\ddot{k}_B = k_B^{**}$. The bank will be indifferent between transacting and not transacting. When $k_B^{**} < \dot{k} \leq k_B^*$, $\ddot{k}_B = \dot{k}$. The bank will operate on the B_1 function and make a strictly positive surplus since if it does not adhere to the minimum cash stipulation and instead operates on the B_2 function, it will make a loss. When $\dot{k} > k_B^*$, $\ddot{k}_B = k_B^*$. The bank will operate on the B_2 function and be indifferent between transacting and not transacting. This completes the proof.

Case (i)

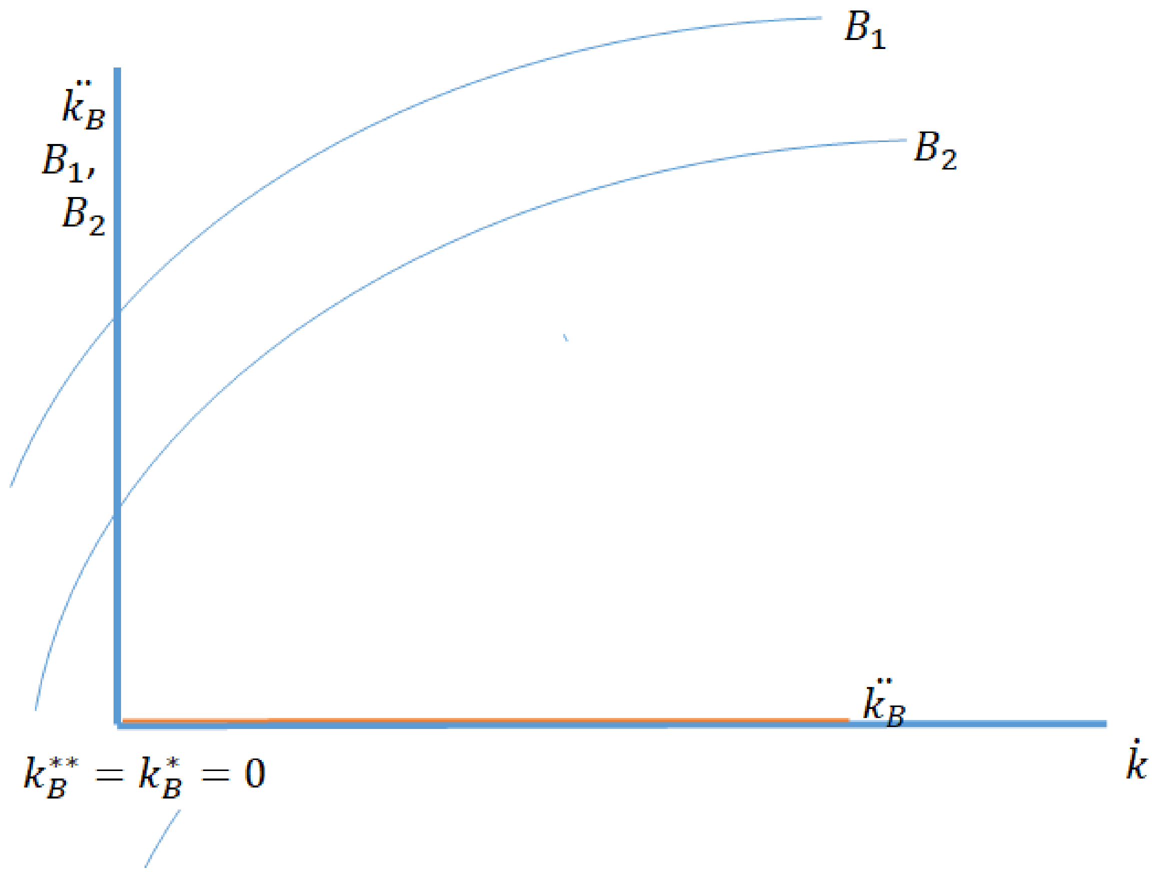

Figure 2: $k_B^{*} = \ k_B^{**} = 0$

Case (ii)

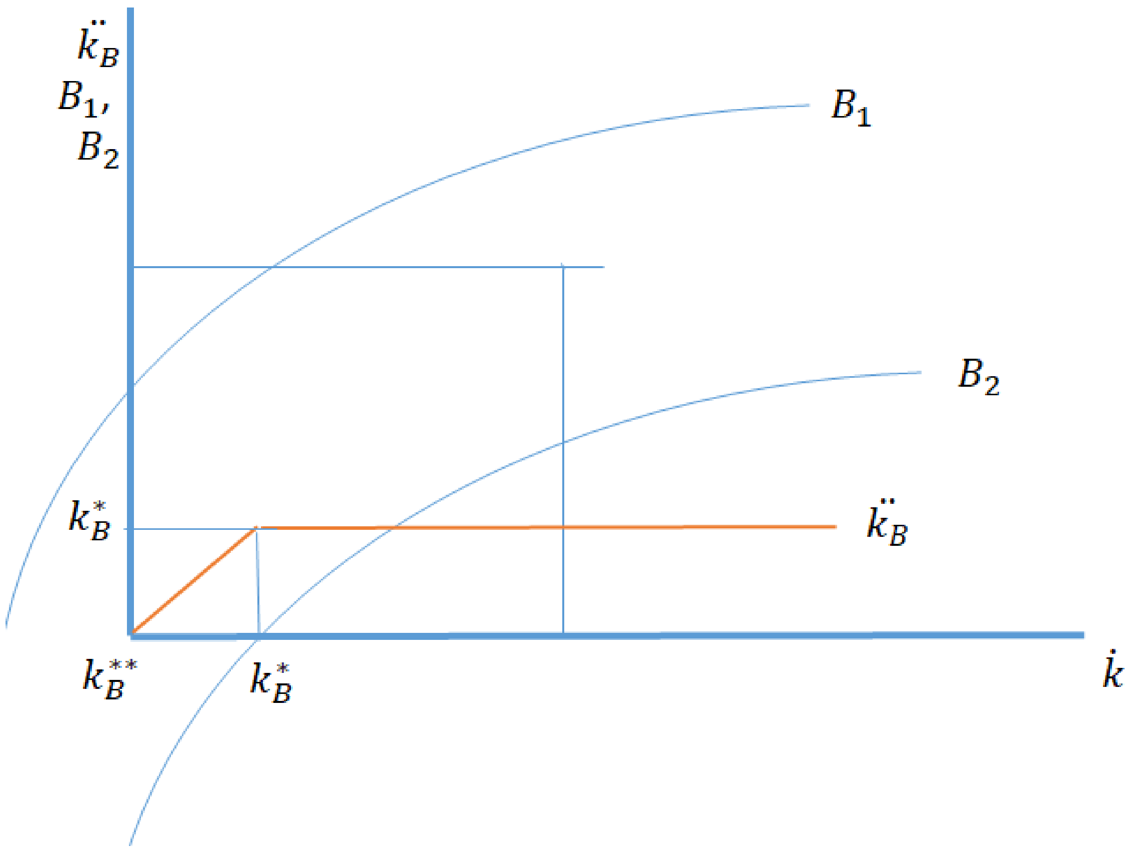

Figure 3: $k_B^* > \quad k_B^{**} = 0$

Case (iii)

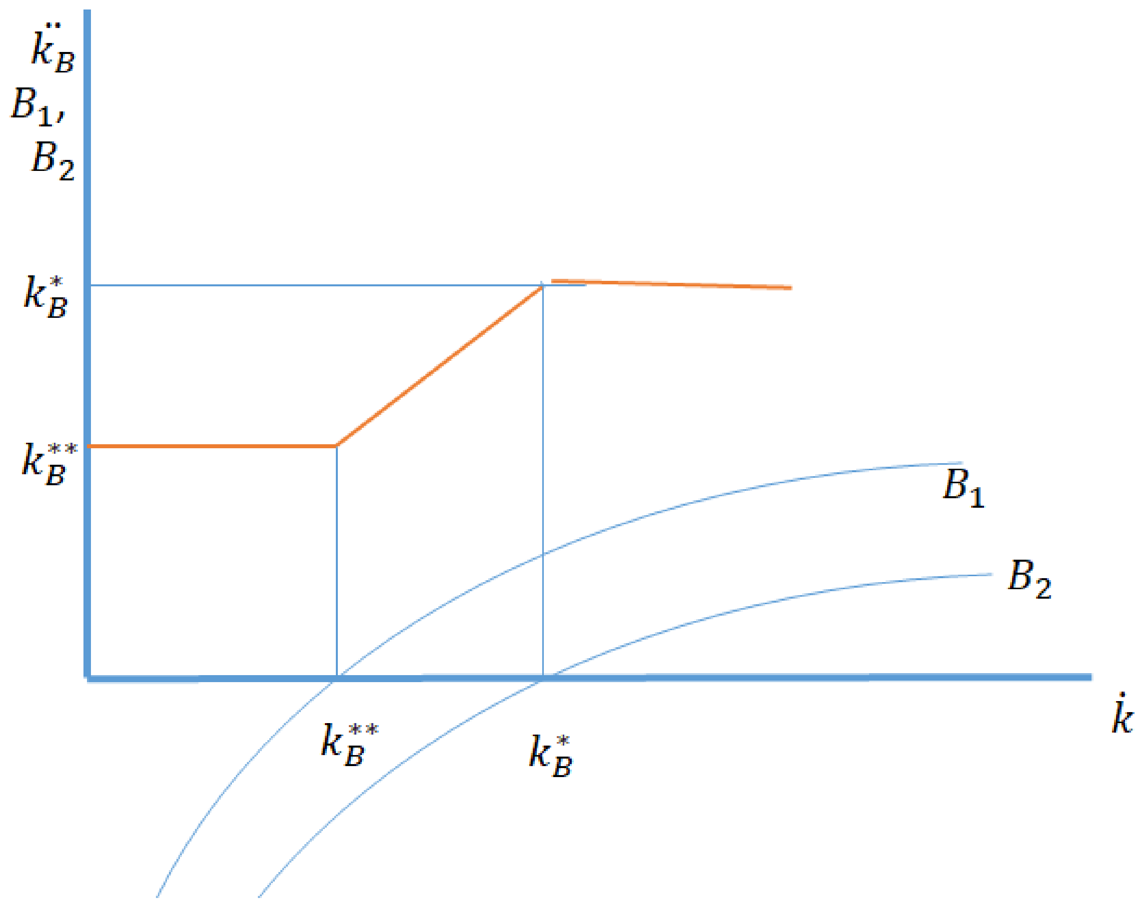

Figure 4: $k_B^* > \ k_B^{**} > 0$.

Lemma 7: *For all* $n \in [0,1]$, *the minimum cash proportion is non-decreasing with an increase in the risk aversion of the bank, the cash value of the deal, and with the management fees; and non-increasing with the premium the ARC is able to earn above the returns from the IBC, and the written down value of debt.*

Proof: Recall that there are three cases under which the minimum cash proportion has to be calculated. The proof can be divided into two broad categories: a. the change in the parameters does not change the type of case. b. the change in the parameters brings about a change in the type of case. We start by looking at changes which fall into category ‘a’.

Suppose the initial equilibrium falls into case(i). Then $\ddot{k}_B = 0$ for all values of $\dot{k} \in [0, 1]$ before and after the change in the parameters.

Suppose the initial equilibrium falls into case(ii). Recall for $0 \leq \dot{k} \leq k_B^*$, $\ddot{k_B} = \dot{k}$, and if $\dot{k} > k_B^*$, $\ddot{k_B} = k_B^*$. If n or f increases or P decreases, both $B_1$and B_2 shift downwards. Let the new thresholds be denoted as $(k_B^{**})_{new}$ and $(k_B^*)_{new}$. Note $(k_B^{**})_{new} = k_B^{**} = 0$, since the case type has remained the same, but $(k_B^*)_{new} > k_B^*$. Thus $(\ddot{k}_B)_{new} = \ddot{k_B} = \dot{k}$ for $\dot{k} \leq k_B^*$ but $(\ddot{k}_B)_{new} > \ddot{k_B}$ for $\dot{k} > k_B^*$. Thus $\ddot{k}_B)_{new} \geq \ddot{k_B}$ for all $\dot{k} \in [0,1]$ with strict inequality for $\dot{k} > k_B^*$. If W or C change, B_2 is not affected hence $(\ddot{k}_B)_{new} = \ddot{k_B}$ for $\dot{k} \in (0,1)$.

Suppose the initial equilibrium falls into case(iii), the conclusion can be arrived at in the same manner as for case (ii), with the only difference being that in this case a change in W or C can change the minimum cash threshold.

Next we look at changes which fall into category 'b', i.e. a change in the parameters leads to a change in the case. Note that $\frac{\partial B_i}{\partial f} = -(1-k)$, $\frac{\partial B_i}{\partial P} = \frac{1}{P^2}$, and $\frac{\partial B_i}{\partial n} = k^n \log \mathrm{k}$, for $i = \{1,2\}$. Thus an increase in f will cause B_i to shift downward at all $k \in [0,1)$ and remain fixed at $k = 1$. A decrease in P will cause B_i to shift downward at all $k \in [0,1]$. At n=0, an increase in n will leave B_i unchanged at $k = 1$ with $B_i(k=1) = \left(1 - \frac{1}{P}\right)$ and cause B_i to shift downward for all $k \in [0, 1)$. For all $n \in (0,1]$, an increase in n will leave B_i unchanged at $k = 0$ and $k = 1$ with $B_i(k=0) = -(f + \frac{1}{P})$, and $B_i(k=1) = \left(1 - \frac{1}{P}\right)$, and cause B_i to shift downward for all $k \in (0,1)$.

From Lemma 2' for $B_1(k=0) \leq 0$, $\frac{\partial k_B^{**}}{\partial C} \geq 0$, and (v) $\frac{\partial k_B^{**}}{\partial W} \leq 0$, and for $B_1(k=0) > 0$, $\frac{\partial k_B^{**}}{\partial C} = 0$, and (v) $\frac{\partial k_B^{**}}{\partial W} = 0$. Thus as W decreases and as C increases, B_1shifts downward or not at all. $W$and C have no impact on B_2.

Hence, as f, or n increase or as P decreases, both $B_1$and B_2 shift downward. As W decreases or C increases B_1shifts downward or not at all, and there is no impact on B_2. Hence, for the abovementioned changes in the parameters it is not possible to start from a 'higher' case (a case denoted by a higher serial number) and move to a 'lower' case (a case denoted by a lower serial number), i.e. one can move from case (i) to case(ii) or to case (iii); and from case (ii) to case (iii) but not in the reverse direction.

If we move from case (i) to case(ii), it must be through a downward shift in $B_1$and B_2through an increase in f or n or a decrease in P. Consequently we move from $\ddot{k_B} = 0$ for all $\dot{k}$ to a case where $\ddot{k_B} = 0$ if $\dot{k} = 0$ and $\ddot{k_B} > 0$ if $\dot{k} > 0$. Hence $(\ddot{k}_B)_{new} \geq \ddot{k_B}$. If we move from case (i) to case(iii) we move from $\ddot{k_B} = 0$ for all $\dot{k}$ to a case where $\ddot{k_B} > 0$ for all $\dot{k} \in [0,1]$. Thus is non-decreasing with respect to $P$and non-increasing with respect to f, n. Note that a change in W or C only affects B_1 and it is not possible to move from case (i) to case(ii) or to case(iii) through a change in W or C.

The move from case(ii) to case(iii) can be understood in a similar fashion.

Corollary 2: *From Lemma 7 we can infer that, ceteris paribus, the minimum cash proportion of the bank is maximized when* $n = 1$. *Similarly, the minimum cash proportion of the bank is minimized when* $min\left\{\frac{W}{C}, 1\right\} = 1$, *i.e.* $W > C$.

4.1. Inefficiency Caused by Structure of Regulatory Rule

Note that there are three instances where the bank is making a strictly positive profit at the minimum cash proportion that would incentivize it to transact: in case (i) , if (i) $\dot{k} > 0$ and (ii) $1 - f - \frac{1}{P} > 0$; In case (ii) if $\dot{k} = 0$, and $B_1(k = 0) > 0$, or if $0 < \dot{k} < k_B^*$; In case(iii), if $k_B^{**} < \dot{k} < k_B^*$.

The possibility that the bank needs to make a supernormal profit in order to transact is a source if inefficiency. Consider the following example. Assume $k_B^* > k_B^{**} > 0$, i.e. we are in case(iii). Let's say that the market starts off at an initial equilibrium given by $\dot{k} = k_B^{**} = k_A^*$. This implies both the ARC and the bank are willing to transact and the market is functioning. Now suppose the regulator raises $\dot{k}$ to a value just under k_B^* . The bank would have to match this higher value of $\dot{k}$ in order to sustain its incentive to transact. If it chooses a lower cash proportion it would be on the B_2 curve and would make losses. If it matches $\dot{k}$, it stays on the B_1 curve and makes a surplus. If it chooses a higher cash proportion it is making an even greater surplus which we do not allow by the definition that $\ddot{k}_B$ is the minimum cash proportion that would incentivize the bank to transact.

But now there is no possibility of a transaction since k_A^* , the maximum cash proportion admissible by the ARC is unchanged. Hence we have one entity, the bank, making a surplus, and the other entity, the ARC, incurring a loss, and no way to incentivize the transaction by transferring some of the surplus from one to the other. This is the outcome of the discontinuity introduced by the central bank's regulatory instrument. Ways of reducing this problem are addressed in the concluding section. We now bootstrap the model by fitting it to the data of the Indian ARC market.

5. Simulation of Market Equilibria in the Indian Market

Developments in the Indian market for secondary debt can be divided into 4 phases:

Table 3: Phases in the Indian Market for Secondary Debt

Phase	Period	Stipulated cash proportion	Size of aggregate NPAs	Age of NPAs	Deal value, C	Volume of transaction	Comments
1	2002-06	None	6% of bank assets	Vintage –low written down value	Low (~ 80% discount on book value)	Medium	Transaction volume made a late surge on account of a large transaction by a public sector supported ARC
2	2006 -13	5%	2.6% of bank assets	Vintage –low written down value	Low (~ 80% discount on book value)	LOW - book value of NPAs sold between 2006-07 and 2012-13 averaged around Rs. 9500 crore annually.	In 2010, central bank mandated ARCs hold 5% of security receipts in trust till all securities were redeemed
3	2013-14	5%	High 4.1% of NPAs	Low - High written down value	Medium (~ 60% discount on book value)	High - Around 40 per cent of the total volume of ARC transactions (in terms of book value) since their inception took place in this short phase of 13 months	RBI eased regulation on ARC market, for instance permitting banks to sell SMA2 accounts
4	2014-present	In August minimum cash proportion raised to 15%. In April 2017 and April	High	Low - High written down value	Medium (60 -70% discount on book value)	Low	The RBI changed the method of calculation of the management fees to the

Phase	Period	Stipulated cash proportion	Size of aggregate NPAs	Age of NPAs	Deal value, C	Volume of transaction	Comments
		2018, this was raised to 50% and 90% respectively					disadvantage of ARCs

Phase 1: 2002-2006

Phase begins with the formulation of the SARFAESI Act 2002 when ARC's were conceptualized. At that time NPAs constituted 6% of bank assets, a sizable number. However, on account of the novelty of the idea, few asset sales took place till 2005 when there was a windfall set of transactions by ARCIL, partly owned by the State Bank of India which is India's largest bank as well as India's largest public sector bank. The cash proportion of the transactions was low as can be inferred from the fact that the central bank's first directive on the minimum cash proportion dated August 2006 set the floor on cash proportion at a miniscule 5%.

IN normal situations, bank credit is available at between 18% and 25%, depending on the assessed risks. Due to the participation of a public sector entity eager to kick start the market we take an interest rate at 20%, close to the lower end of the spectrum. Hence Z =1.2. As explained earlier, $C = bvs_I$. Given the absence of a bankruptcy framework, we assume $s_I = L$, the liquidation value. Banks earned close to 25-30% of the book value of an asset through liquidation but an average case took about 6 years and the time value of money also needs to be taken into account. Hence we set $s_I = 15\% = 0.15$. Since the NPAs were of significant maturity, the written down value was close to zero (note this means that we only need to consider B_2 when assessing the incentives of the bank). The absence of a secondary market in security receipts inhibited v which we assume is equal to 1. As shown by the data, the deal value $C = 20\%$ of book value. This implies $P = b \cong 1.3$. The management fee is taken as f =0.05, i.e. 5%, although for larger deals it would be as low at 1.5%. The cash proportion at which transactions took place is assumed to be 5%. Based on these values, the bank would be willing to transact at a 5% cash proportion only if

$B \equiv B_2 = k^n - (1-k)(0.05) - \frac{1}{1.3} \geq 0$. This gives n = 0.0675. The ARC would be willing to give 5% cash only if

$$A' = k^m - (1.2 + .05)k + .05 \geq 0$$

This gives m = 1.463. Note n =0.0675 implies a high level of bullishness on the part of the bank about security receipts since the bank expects full recovery from security receipts when n= 0. This explains the low level of transactions. The windfall transaction by the public sector bank can be regarded as a market creation investment initiated by a public sector entity in the public interest.

Phase 2: 2006 – 2013

In this period the economy was doing well on account of both the global commodities cycle and domestic factors. However, banks were bearish on security receipts – on account of their past experience. Nevertheless data shows (see Table 1 Bhagwati et al 2017) that the average deal value increased to around 25% of the book value. This could be on account of a marginal decrease in maturities of NPAs resulting in higher written down values.

Table 4: Cumulative ARC Business in India Rs. Crores.

year	**2007**	**2008**	**2009**	**2010**	**2011**	**2012**	**2013**	**2014**	**2015**
book value	28544	41414	51542	62217	74088	80500	139700	190600	210600
C	7436	10658	12801	14051	15859	18900	39310	61740	75830

Source: Bhagwati et al 2017

We assume liquidation value also increases on account of the good performance of the economy to just above 20% (this would be consistence with a liquidation value of 35% adjusted for the time taken for liquidation which remained roughly the same as in Phase 1). As a result, the premium, P, created by the ARC reduces to 1.2. The interest rate continues to be Z =1.2; the management fees continues to be f =0.05. We assume that on account of the good performance of the economy, new NPAs are not generated, and the written down value of NPAs is zero. Based on these values, assuming the risk aversion coefficient n stays at 0.065, the bank would be willing to transact only if

$B \equiv k^{.065} - (1-k)(0.05) - \frac{1}{1.2} \geq 0$. This gives $k_B^* = \sim 0.12$. The ARC would be willing to give 12% cash only if

$$A' = (0.12)^m - (1.2 + .05)(0.12) + .05 \geq 0$$

This gives $m \cong 1.09$. Thus assuming the risk aversion of the bank remains constant, the risk aversion of the ARC has to decline by 25% in order to make transactions possible. In fact, the poor past performance of security receipts could well justify an increase in the risk aversion of banks, thus calling for an even lower level of risk aversion by the ARC. This lower level of risk aversion on the part of the ARC was required even as the central bank was taking steps that made it harder for ARCs to turn a profit, for instance, removing the priority of ARCs with respect of the return from the security receipts. This explains the low level of transactions in this phase.

Phase 3: 2013 – 2014

By 2008 the global economy has undergone a deep recession, and by 2013 the domestic economy was also slowing down. NPAs amounted to 4.1% of the market, a 1.5% increase from the previous phase. These included a large number of newly generated NPAs. Thus the NPAs had a high written down value. This created an opportunity for significant accounting benefits to accrue to the bank from a transaction with the ARC provided the minimum cash ratio was satisfied. The minimum cash ratio remained at 5%. The central bank encouraged early sale of stressed assets by a number of measures including permitting ARCs to buy special mention accounts that were in the final stages of stress prior to being declared as NPAs. The deal size increased to about 40% of book value. We assume this was entirely on account of the increase in the written down value of the NPAs and not on account of a change in P or s_I.

The following parameter values remain unchanged from Phase 2: Z = 1.2; f=.05, $\dot{k} = 0.05$, P = 1.2. We assume that n = .065, as before, and that the average written down value of NPAs was 30% of the book value or 0.3. Since the deal value was 40% of the book value, $\frac{W}{C} = \frac{0.3}{0.4} = 0.75$.

This gives

$B_1 \equiv k^{0.065} - (1-k)(0.05) - \frac{1}{1.2} + 0.75 \geq 0$ if $k \geq 0.05$, and

$B_2 \equiv k^{0.065} - (1-k)(0.05) - \frac{1}{1.2} \geq 0$ if $k < 0.05$

The availability of accounting gains meant that provided the regulatory stipulation $k \geq 0.05$ was satisfied, banks were willing to transact at a cash proportion which is greater than zero by a miniscule amount, i.e. $k_B'' = k_B^{**}$ is positive but less than 0.05, the stipulated minimum. If the regulatory stipulation was not satisfied, banks would transact at $k \geq 0.12$, i.e. $k_B^* = 0.12$. This means $k_B^* > \dot{k} > k_B^{**} > 0$, and we are in case (iii) of the model incorporating accounting benefits. In this situation and $\ddot{k}_B = \dot{k} = 0.05$, i.e. the bank will enforce the regulator's directive and make a surplus. The ARC would be willing to give 5% cash only if

$$A' = (0.05)^m - (1.2 + .05)(0.05) + .05 \geq 0$$

This implies $m \leq 1.46$, a moderate level of risk aversion on the part of the ARC. Thus we see that transactions become feasible in Phase 3. This explains the sharp increase in volumes.

Phase 4: 2014 onwards

This phase was characteristic by continued weakness in economic conditions, along with progressive increases in the volume of cash recommended by the central bank. The central bank raised the ARC investment requirement from 5 per cent to 15 per cent in August 2014. From April 2017, if a bank purchased security receipts on assets it has offloaded to an ARC beyond 50 percent of the marked-down value of the asset, then it would have to set aside capital as if it were an NPA. From April 1, 2018, this threshold of 50 percent was reduced to 10 percent. This phase was marked by a significant slowdown of activity in the distressed debt market. To understand the developments let us apply the model with the same parameters as used in the analysis of Phase 3, except with $\dot{k}$, the minimum cash requirement stipulated by the central bank having increased from 5% to 15% and then from 15% to 90%.

For the bank

$B_1 \equiv k^{0.065} - (1-k)(0.05) - \frac{1}{1.2} + 0.75 \geq 0$ if $k \geq 0.15$, and

$B_2 \equiv k^{0.065} - (1-k)(0.05) - \frac{1}{1.2} \geq 0$ if $k < 0.15$

Recall k_B^{**} is positive but close to zero, and $k_B^* = 0.12 < \dot{k} = 0.15$ Hence, as per the analysis of case (iii), $\ddot{k}_B$, the threshold cash proportion below which the bank will not transact, is 0.12.

The ARC would be willing to give 12% cash only if

$$A' = (0.12)^m - (1.2 + .05)(0.12) + .05 \geq 0$$

This implies $m \leq 1.09$ which implies a very low level of risk aversion on the part of the ARC. Note the bank is on the B_1curve and would be making significant profits were a transaction to take place. Under the circumstances the central bank could have attempted to trigger the market by transferring some surplus from the bank to the ARC, for instance, by allowing the ARC to claim seniority in the earnings accruing from security receipts.

If $\dot{k} = 0.9$, then $\dot{k} > k_B^* > k_B^{**}$. Hence $\ddot{k}_B = k_B^* = 0.12$.

The ARC would be willing to give 90% cash only if

$$A' = (0.9)^m - (1.2 + .05)(0.9) + .05 \geq 0$$

This implies $m \leq -0.68$, i.e the ARC has to believe it will realize more than the value of the security receipts it possesses in order to transact(recall when m=1, the ARC believes it will recover the exact market value). Hence our model predicts a very low level of transaction in the market for secondary debt given this regulatory stipulation. There is also no possibility of salvaging the market by transferring surplus as in the case where $\dot{k} = 0.15$.

6. Conclusion

The model identifies the risk aversion of the bank and the ARC, the value created by the ARC, the management fees charged for managing the security receipts held by the bank, the cost of capital of the ARC, and the minimum cash proportion stipulated by the central bank as important determinants of market viability. It establishes that banks always want all-cash deals while ARCs always want zero-cash deals. For all admissible values of the parameters it establishes that there exists a minimum cash proportion that would incentivize the bank and a maximum cash proportion that would incentivize the ARC to transact. It establishes the direction of change of these threshold values with changes in the values of the relevant parameters. It thus enables market simulations to explore the existence of gains to trade and policy development to establish the market on a sound financial footing.

For specific values of the parameters it identifies sufficient conditions for the feasibility of a transaction. It establishes that if the risk aversion of banks increases as the economy moves from the crest to the trough of a business cycle, the risk aversion of the ARC must decrease in order for a transaction to take place. It shows that for certain values of the minimum cash proportion stipulated by the central bank the market could become infeasible as the ARC is not breaking even, although the bank is making positive profits. The model is able to satisfactorily explain the data from the ARC market in India since 2002 and show that the current minimum cash proportion stipulated by the central bank renders the market unviable.

Note that the value of the deal, $C = bvs_I$. This implies that all the value created by the ARC is entirely captured by the bank, a situation which could emerge if there are a few banks selling NPAs to a highly competitive ARC market. In India, 80% of the NPAs are held by the public sector banks, and the number of ARCs has increased to 29. Hence the assumption may be valid. If we assume the value created by the ARC is split between the bank and the ARC, the minimum threshold of the bank and the maximum threshold of the ARC would both increase. Overall, there would not be much material change in the conditions required to make a transactions feasible, although the algebraic complexity of the computations would certainly increase.

IN terms of the broad contours of the market, an important insight relates to the fact that ARCs need to behave counter-cyclically when banks behave pro-cyclically. This implies that the market for distressed debt becomes more unviable at the trough of the business cycle, which is precisely the time one needs the market to function at its best. Broadly speaking , incentivizing banks to behave less pro-cyclically would help in two ways – it would lead to the generation of lower volumes of NPAs at the trough, and would allow ARCs to be more risk averse at the trough. This overarching requirement needs to be kept in mind as we discuss specific measures that would trigger the market.

Based on the model the ARC market can be stimulated by increasing P, the value created by the ARC; reducing Z, the ARC's cost of capital; and optimizing $\dot{k}$, the minimum cash proportion stipulated by the central bank. A change in the management fees f has opposite effects on the incentives of the bank and the ARC. We discuss each of these variables in turn.

Increasing P: P is composed of b, the bankruptcy avoidance premium, and the value created by the ARC through securitization. A perverse way of increasing P is to preserve a malfunctioning bankruptcy landscape, hence keeping b high. However b can also be increased in genuinely healthful ways. For instance, ARCs can be incentivized to engage in asset reconstruction. In India, the central bank changed regulation to allow the ARC to develop an equity position in the borrower company to incentivize asset reconstruction. There needs to be further improvement in the legal infrastructure concerning recovery/ business reorganization.

The creation of value through securitization depends on the development of a secondary market in distressed debt. This in turn depends on allowing new market players to enter as sellers as well as buyers of security receipts. However when making such allowances the regulator should keep the viability of the market for new and existing players in mind.

Reducing Z: Allowing ARCs to invest in assets in the pre-NPA phase would improve their business potential and thereby increase the supply of funds to ARCs. In India, ARCs are only allowed to buy NPAs (they cannot buy stressed assets not declared as NPAs). Further they can only provide interim finance or 'rescue funding' to companies whose NPAs they own. This prevents them from contributing interim finance prior to the classification of an account as an NPA. Their participation at that stage, for instance at early stages of the stressed asset lifecycle, could allow give stressed companies access to vital funds, and prevent the onset of bankruptcy, an event that reduces the value of a company in many ways. Further, the priority of interim financiers in a bankruptcy case could be extended to interim financiers in stressed assets as well.

Optimizing $\dot{k}$: The highest level of $\dot{k}$ that would be acceptable to an ARC is given by $\frac{f}{Z+f-1}$. Any $\dot{k}$ above this level would result in an infeasible market unless ARCs are being given certain indirect benefits, for instance through the tax structure. If $\frac{\frac{1}{p}+f}{1+f} < \dot{k} < 1$, the bank would be willing to transact no matter how risk averse it may be. Further, there is an important point related to the structure of the regulatory rule.

We saw in the analysis of Phase (iv) of the market that a regulatory rule that allows significant accounting benefits from the sale of NPAs if the cash proportion is above a certain threshold and no benefits otherwise, can lead to an infeasible market even though one party has a surplus that could be transferred to the other party.

The problem could be reduced (though not eliminated) by modifying the rule as follows: if the cash proportion is greater than or equal to $\dot{k}$, the bank makes an accounting profit of $\min\{W, C\}$, the minimum of the written down value and the deal value. Otherwise, its profit is restricted to $k \,.\min\{W, C\}$ where k is the actual cash proportion. This reduces the discontinuity in the bank's surplus functions and thereby reduces the unused surplus in the system.

IN our model, we assume that the risk aversion parameter of the bank, $n \in [0,1]$, and the risk aversion parameter of the ARC, $m \in [1,\infty)$. In particular, this assumption rules out the possibility that either the bank or the ARC could ascribe a value to security receipts that is greater than their book value. Given the objective of the model to analyze the market for distressed debt in a difficult business environment, this assumption seems appropriate.

Given the high level of NPAs in many jurisdictions across the globe, all measures to enable expeditious recovery of dues to the maximum extent possible must be encouraged. The support of the ARCs is a critical portion of the slew of measures that need to be put in place.

7. References

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